

# Machine Organization and ALP

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## Chapter 1 – Intel Machine Architecture

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# Contents

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- ✧ Introduction to Microprocessors
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- ✧ Programmers' View 80x86 Systems

# Introduction to Microprocessors

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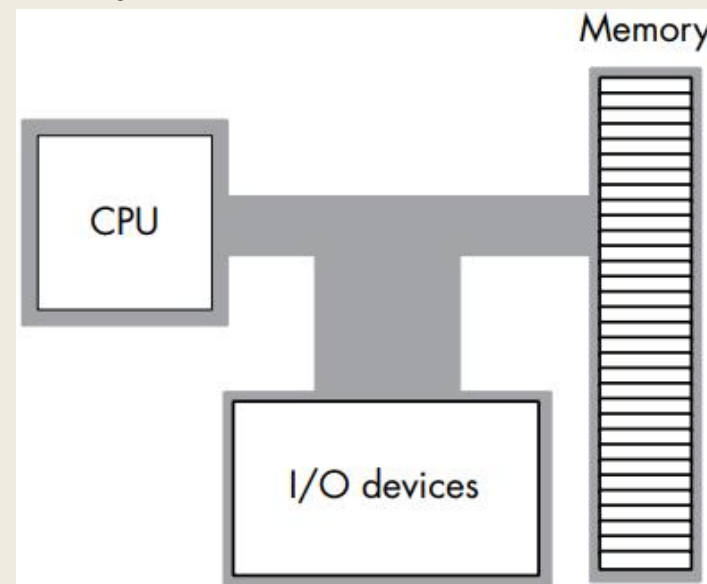


- ✧ The heart of any computer is the Central Processing Unit
- ✧ It is usually a single piece of high-density circuitry built on single integrated circuit (IC) silicon chip
- ✧ The Intel CPU family is generally classified as a von Neumann architecture machine.
- ✧ Von Neumann computer systems contain three main building blocks: the central processing unit (CPU), memory, and input/output (I/O) devices.
- ✧ These three components are interconnected via the system bus (consisting of the address, data, and control buses)



# Introduction to Microprocessors

- ✧ Microprocessor is a programmable, multipurpose, clock -driven, register-based electronic device
  - Reads binary instructions from a storage device called memory
  - Accepts binary data as input and processes data according to those instructions and provides results as output
- ✧ Generally, Microprocessor is a digital device on a chip that fetch instructions from memory, decode and execute them





# Basic Terms in Microprocessors

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- ✧ Instruction Set – The group of commands that the microprocessor can understand
  - It is an interface between hardware and software.
- ✧ Bus – Set of conductors intended to transmit data, address or control information to different elements in a microprocessor.
  - A microprocessor will have three types of buses, i.e., data bus, address bus, and control bus.
- ✧ IPC (Instructions Per Cycle) – It is a measure of how many instructions a CPU is capable of executing in a single clock.
- ✧ Clock Speed – The number of operations per second the processor can perform.
  - Measured in Hertz – example in MHz or GHz



# Basic Terms in Microprocessors

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- ✧ Bandwidth – The number of bits processed in a single instruction
- ✧ Word Length – The number of bits the processor can process at a time is called the word length of the processor
  - 8-bit Microprocessor may process 8 -bit data at a time.
  - The range of word length is from 4 bits to 64 bits depending upon the type of the microcomputer.
- ✧ Data Types – The microprocessor supports multiple data type formats like binary, ASCII, signed & unsigned numbers
- ✧ Cache Memory – It is a random-access memory integrated into the processor to access data more quickly than from a regular RAM.



# Features of Microprocessor

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- ✧ Low Cost – Due to integrated circuit technology microprocessors are available at very low cost.
  - It will reduce the cost of a computer system.
- ✧ High Speed – Due to the technology involved in it, the microprocessor can work at very high speed.
  - It can execute millions or billions of instructions per second
- ✧ Small Size – A microprocessor is fabricated in a very less footprint due to very large scale and ultra large scale integration technology.
  - Because of this, the size of the computer system is reduced.
- ✧ Versatile – The same chip can be used for several applications, therefore, microprocessors are versatile.



# Features of Microprocessor

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- ✧ Low Power Consumption – Microprocessors are using metal oxide semiconductor technology, which consumes less power.
- ✧ Less Heat Generation – Microprocessors uses semiconductor technology which will not emit much heat as compared to vacuum tube devices.
- ✧ Reliable – Since microprocessors use semiconductor technology, therefore, the failure rate is very less. Hence it is very reliable.
- ✧ Portable – Due to the small size and low power consumption microprocessors are portable.



# Evolution of Intel Microprocessor

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## ✧ *4 Bit Microprocessors:*

### ✧ Intel 4004:

- The world's first microprocessor – invented at 1971
- It was a 4-bit microprocessor – programmable controller on a chip
- It addressed a mere 4096, 4-bit-wide memory locations
- It had 2,300 transistors
- The 4004 instruction set contained only 45 instructions
- Execute instructions at the rate of 50 – 60 KIPs (kilo-instructions per second)
- Its clock speed was 740KHz
- This was slow when compared to the 100,000 instructions executed per second by the 30-ton ENIAC computer in 1946.
- The difference was that the 4004 weighed much less than an ounce



# Evolution of Intel Microprocessor

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## ✧ *4 Bit Microprocessors:*

### ✧ Intel 4040:

- An updated version of the earlier intel 4004 microprocessor
- The 4040 operated at a higher speed, although it lacked improvements in word width and memory size
- 8 KB of program memory
- 640 bytes of addressable memory
- Could execute approximately 60,000 instructions per second
- It had 3000 transistors
- Clock speed is between 500 kHz and 740 kHz
- Uses a crystal to 5185 MHz



# Evolution of Intel Microprocessor

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## ✧ *8 bit Microprocessors:*

### ✧ Intel 8008:

- An extended 8-bit version of the 4004 microprocessor
- The 8008 addressed an expanded memory size (16K bytes) with 14 bit address bus
- Contained additional instructions (a total of 48) that provided an opportunity for its application in more advanced systems
- It was first 8-bit microprocessor
- Its clock speed was 500 KHz.
- Could execute less than 36,000 to 80,000 at 0.8 MHz instructions per second
- Its somewhat small memory size, slow speed, and instruction set limited its usefulness



# Evolution of Intel Microprocessor

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## ✧ *8 bit Microprocessors:*

### ✧ Intel 8080:

- Address more memory and execute additional instructions
- Execute instructions 10 times faster than the intel 8008
- 500,000 instructions per second
- The 8080 also addressed four times more memory (64K bytes) than the 8008 (16K bytes)



# Evolution of Intel Microprocessor

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## ✧ *8 bit Microprocessors:*

### ✧ Intel 8085:

- Updated version of the 8080
- The last 8-bit, general-purpose microprocessor developed by Intel
- 769,230 instructions per second
- It had 246 instructions
- It had 6,500 transistors
- The main advantages of the 8085 were its internal clock generator, internal system controller, and higher clock frequency
- This higher level of component integration reduced the 8085's cost and increased its usefulness.



# Evolution of Intel Microprocessor

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## ✧ *16 bit Microprocessors:*

### ✧ Intel 8086:

- It was first 16-bit microprocessor
- Addressed 1M byte of memory, which was 16 times more memory than the 8085
- It had 29,000 transistors
- Could execute 2.5 million instructions per second
- It had 22,000 instructions
- Its clock speed is 4.77 MHz, 8 MHz and 10 MHz, depending on the version
- It had Multiply and Divide instructions



# Evolution of Intel Microprocessor

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## ✧ *16 bit Microprocessors:*

### ✧ Intel 8088:

- It was a 16-bit microprocessor
- It was created as a cheaper version of Intel's 8086
- It was a 16-bit processor with an 8-bit external bus
- Could execute 2.5 million instructions per second



# Evolution of Intel Microprocessor

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## ✧ *16 bit Microprocessors:*

### ✧ Intel 80186 and 80188:

- They were 16-bit microprocessor
- Clock speed was 6 MHz
- 80188 was a cheaper version of 80186 with an 8 – bit external data bus
- They had additional components like:
  - Interrupt Controller
  - Clock Generator
  - Local Bus Controller
  - Counters



# Evolution of Intel Microprocessor

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## ✧ *32 bit Microprocessors:*

### ✧ Intel 80386:

- Intel's first practical 32-bit microprocessor
- Contained a 32-bit data bus and a 32-bit memory address
- Addressed up to 4G bytes of memory
- The 80386 was available in a few modified versions:
  - 80386SX which addressed 16M bytes of memory through a 16-bit data and 24-bit address bus
  - The 80386SL/80386SLC, which addressed 32M bytes of memory through a 16-bit data and 25-bit address bus
  - An 80386SLC version contained an internal cache memory that allowed it to process data at even higher rates



# Evolution of Intel Microprocessor

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## ✧ *32 bit Microprocessors:*

### ✧ Intel 80486:

- It had 1.2 million transistors
- Its clock speed varied from 6 MHz to 100 MHz depending upon the various versions.
- It had five different versions:
  - 80486 DX, 80486 SX, 80486 DX2, 80486 SL, 80486 DX4
- Incorporated an 80386-like microprocessor, an 80387-like numeric coprocessor, and an 8K-byte cache memory system into one integrated package



# Evolution of Intel Microprocessor

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## ✧ *32 bit Microprocessors:*

### ✧ The Pentium Microprocessor:

- Originally labeled the P5 or 80586
- Operated with a clocking frequency of 60 MHz and 66 MHz
- Could execute 110 million instructions per second
- The double-clocked Pentium, operating at 120 MHz and 133 MHz, was also available, as were higher-speed versions
- Cache size was increased to 16K bytes from the 8K
- Contained an 8K-byte instruction cache and an 8K-byte data cache



# Evolution of Intel Microprocessor

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## ✧ *32 bit Microprocessors:*

### ✧ Pentium Pro Processor:

- Contains 21 million transistors, integer units, as well as a floating-point unit to increase the performance of most software
- The basic clock frequency was 150 MHz and 166 MHz in the initial offering made available in late 1995.
- In addition to the internal 16K level-one (L1) cache it contains a 256K level-two (L2) cache
- Uses three execution engines, so it can execute up to three instructions at a time, which can conflict and still execute in parallel.
- It was primarily used in server system



# Evolution of Intel Microprocessor

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## ✧ *32 bit Microprocessors:*

### ✧ Pentium II and Pentium Xeon Microprocessors:

- The Pentium II microprocessor represents a new direction for Intel
  - Instead of being an integrated circuit, it has been placed on a small circuit board.
  - The reason for the change is that the L2 cache found on the main circuit board was not fast enough to function properly with the Pentium II
- Its clock speed was 233 MHz to 500 MHz
- Could execute 333 million instructions per second
- L2 cache & processor were on one circuit



# Evolution of Intel Microprocessor

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## ✧ *32 bit Microprocessors:*

### ✧ Pentium III Microprocessor:

- The Pentium III microprocessor uses a faster core than the Pentium II, but it is still a P6 or Pentium Pro processor.
- It is also available in the slot 1 version mounted on a plastic cartridge and a socket 370 version called a flip-chip
- The slot 1 version contains a 512K cache and the flip-chip version contains a 256K cache
- Both versions use a memory bus speed of 100 MHz
- Available with clock frequencies of up to 1 GHz
- It had 9.5 million transistors



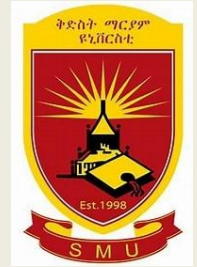
# Evolution of Intel Microprocessor

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## ✧ *32 bit Microprocessors:*

### ✧ Pentium 4 and Core2 Microprocessors:

- The most recent version of the Pentium is called the Core2 by Intel
- The Pentium 4 and Core2 microprocessors use the Intel P6 architecture
- Pentium 4 is available in speeds to 3.2 GHz and faster
- The chip sets that support the Pentium 4 use the RAMBUS or DDR memory technologies
- The Core2 is available at speeds of up to 3 GHz.
- These higher microprocessor speeds are made available by an improvement in the size of the internal integration



# Evolution of Intel Microprocessor

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## ✧ *64 bit Microprocessors:*

### ✧ Pentium 4 & Core2, 64-bit & Multiple Core Microprocessors:

- The 64-bit modification allows the microprocessor to address more than 4G bytes of memory through a wider 64-bit address
- Addresses up to 1T (terabytes) of memory to be accessed
  - It also allows 64-bit integer arithmetic
- The biggest advancement is the inclusion of multiple cores
  - Each core executes a separate task in a program, which increases the speed of execution if a program is written to take advantage of the multiple cores
  - Programs that do this are called multithreaded applications.



# Evolution of Intel Microprocessor

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## ✧ *64 bit Microprocessors:*

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  - Programs that do this are called multithreaded applications.
- Intel Multiple Core technologies include core i3, i5, i7 and i9.
  - *Read about the memory locations addressed by each core technologies.*
  - *Read about other features such as Data Bus, Clock Speed, IPC, number of cores, caches and etc.*



# Basic Architecture of 8086 and 8088 Microprocessors

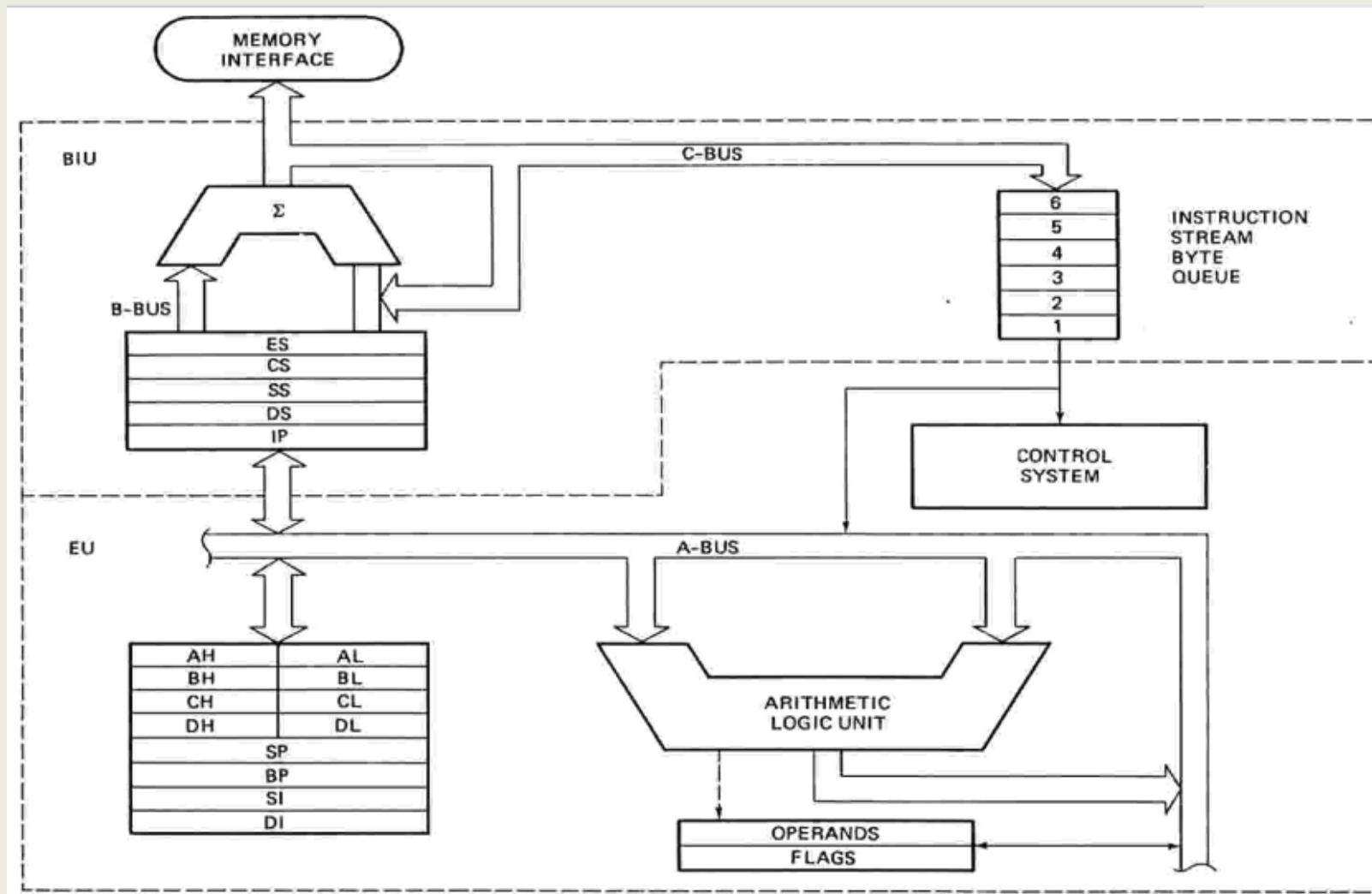
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## ✧ Features of 8086 Microprocessor:

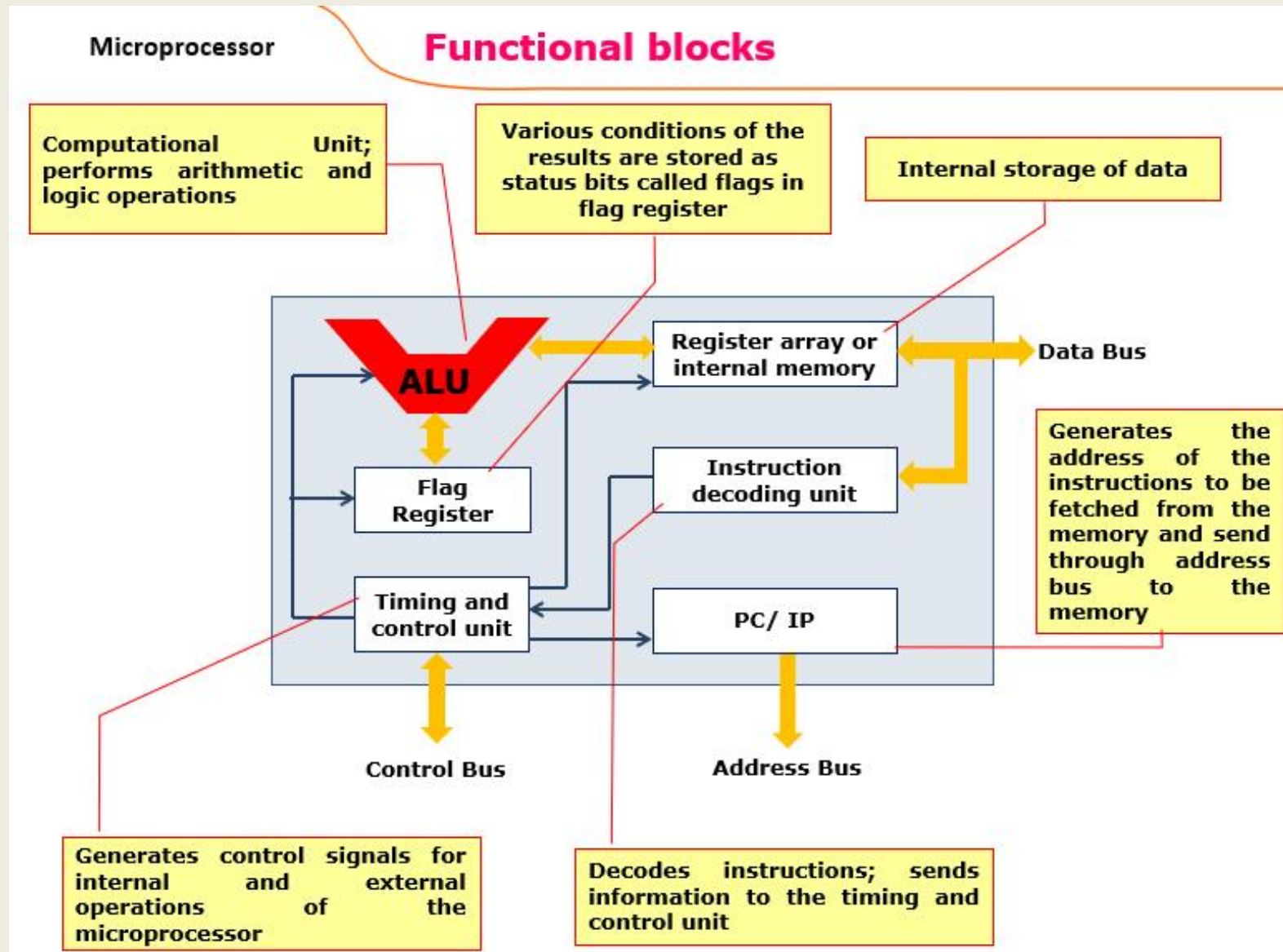
- It is a 16-bit microprocessor and has a 20 bit address bus
- Can access up to  $2^{20}$  memory locations (1MB)
- It can support up to 64K I/O ports and provides 4, 16 -bit registers
- Word size is 16 bits and double word size is 4 bytes
- It has multiplexed address and data bus AD0- AD15 and A16 – A19
- 8086 is designed to operate in two modes, Minimum and Maximum
- It can prefetches up to 6 instruction bytes from memory and queues them in order to speed up instruction execution.
- A 40 pin dual in line package
- Address ranges from 00000H to FFFFFH
- The programming model of the 8086 through the Core2 is considered to be program visible because its registers are used during application programming and are specified by the instructions

# Basic Architecture of 8086 and 8088 Microprocessors

✧ Block diagram of 8086:

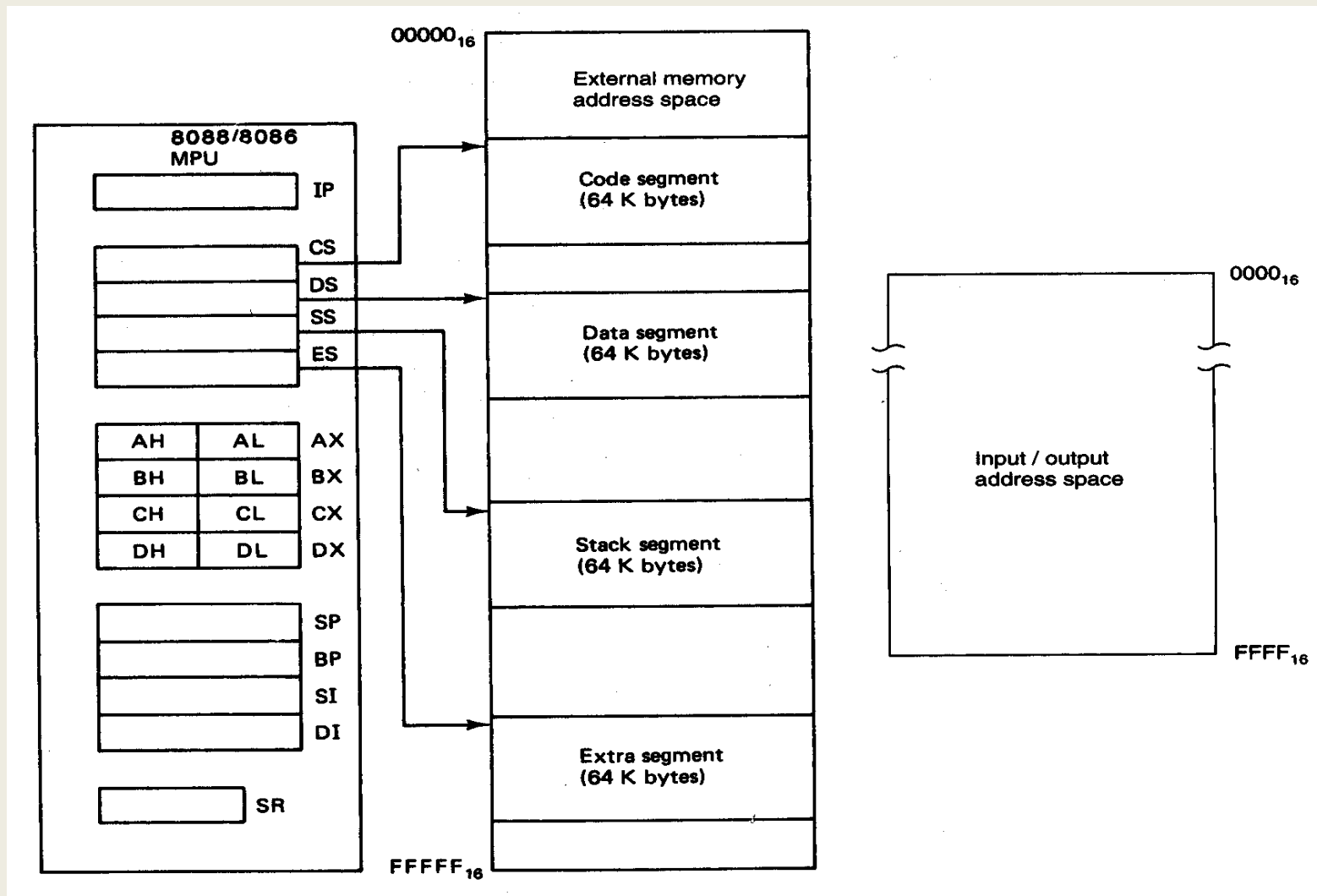


# Basic Architecture of 8086 and 8088 Microprocessors



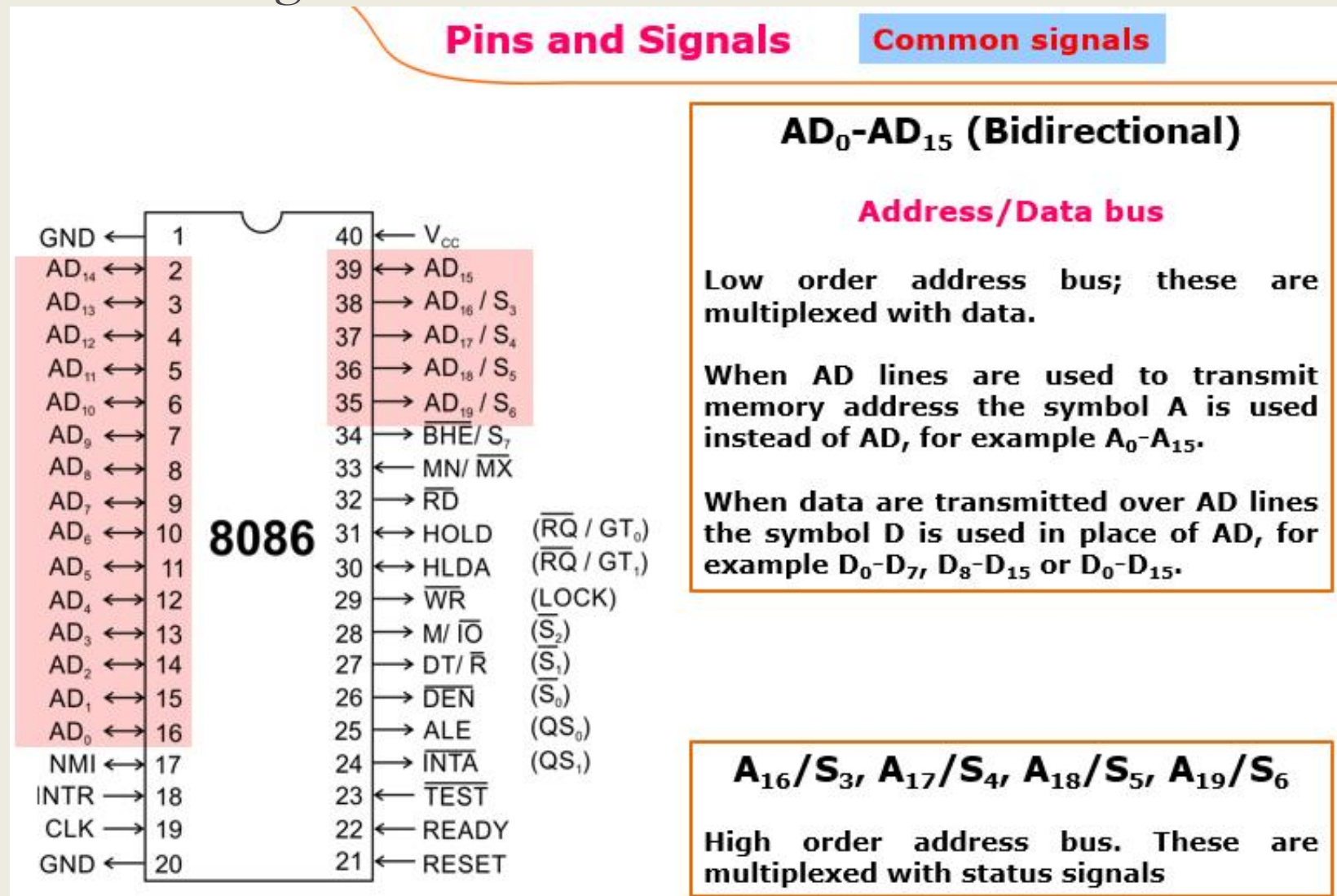
# Basic Architecture of 8086 and 8088 Microprocessors

## ✧ Software Model of the 8086 Microprocessors



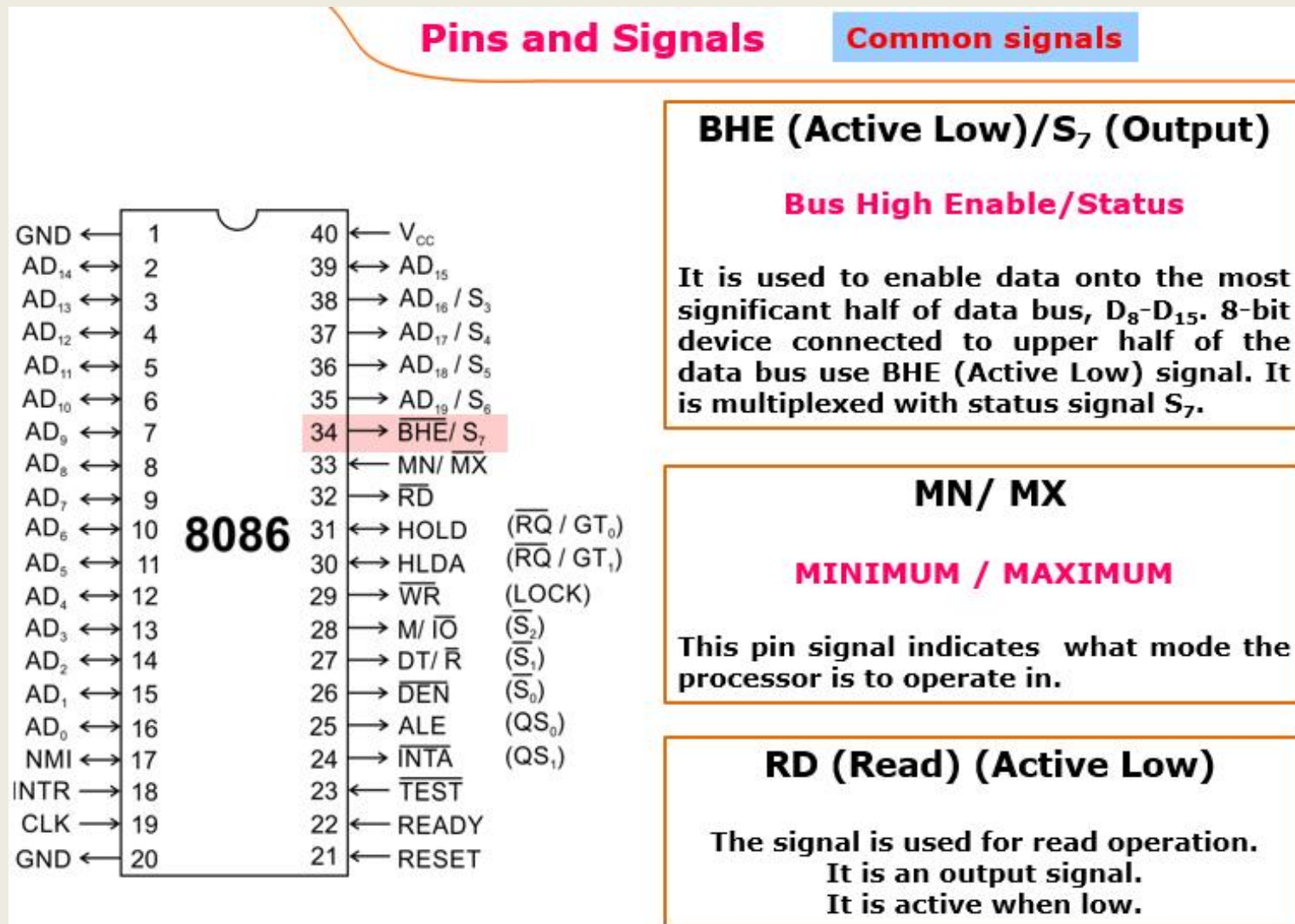
# Basic Architecture of 8086 and 8088 Microprocessors

## ✧ Pins and Signals



# Basic Architecture of 8086 and 8088 Microprocessors

## ✧ Pins and Signals



# Basic Architecture of 8086 and 8088 Microprocessors

## ✧ Pins and Signals

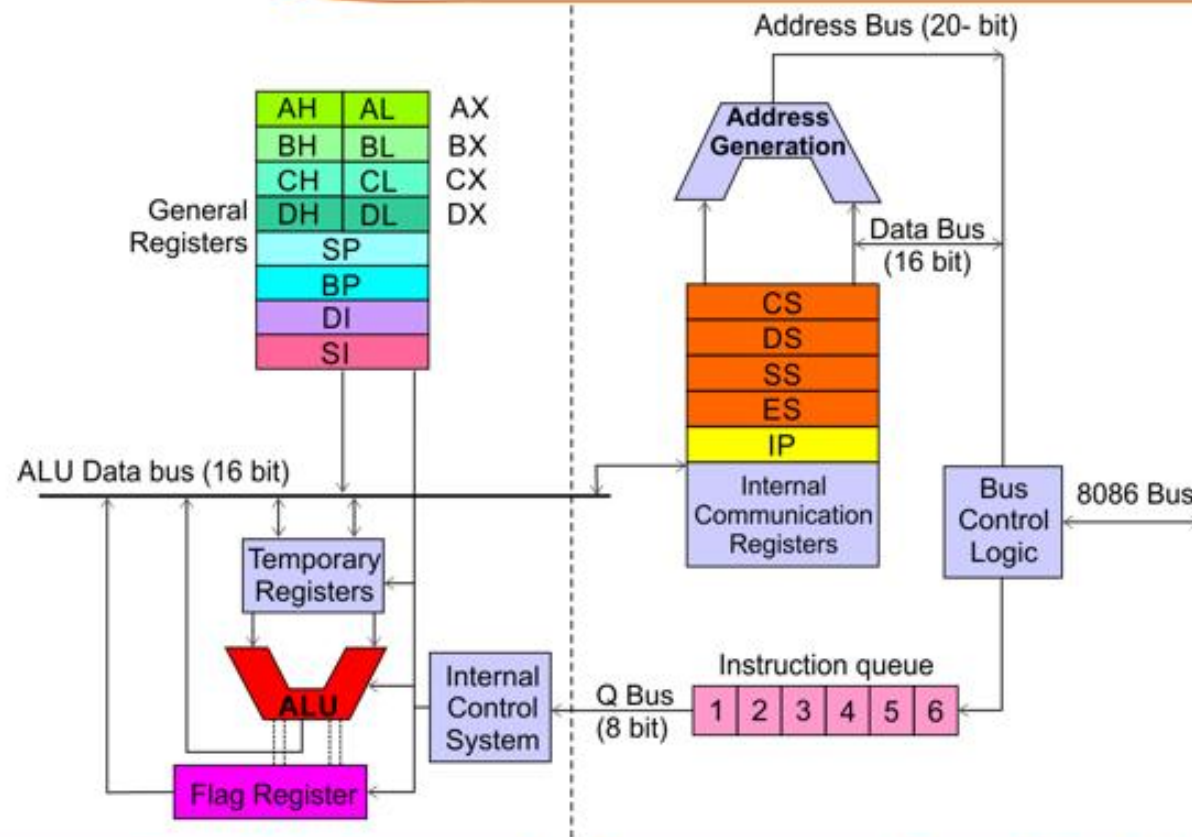
| Pins and Signals                                                                                                                                                 |  | Min/ Max Pins |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------|
| <p>The 8086 microprocessor can work in two modes of operations : <b>Minimum mode</b> and <b>Maximum mode</b>.</p>                                                |  |               |
| <p>In the <u>minimum mode</u> of operation the microprocessor <u>do not</u> associate with any co-processors and can not be used for multiprocessor systems.</p> |  |               |
| <p>In the <u>maximum mode</u> the 8086 <u>can work</u> in multi-processor or co-processor configuration.</p>                                                     |  |               |
| <p>Minimum or maximum mode operations are decided by the pin MN/ MX(Active low).</p>                                                                             |  |               |
| <p>When this pin is <u>high</u> 8086 operates in <u>minimum mode</u> otherwise it operates in <b>Maximum mode</b>.</p>                                           |  |               |

| 8086 Pinout          |                                        |
|----------------------|----------------------------------------|
| GND ← 1              | 40 ← V <sub>CC</sub>                   |
| AD <sub>14</sub> ↔ 2 | 39 ↔ AD <sub>15</sub>                  |
| AD <sub>13</sub> ↔ 3 | 38 ↔ AD <sub>16</sub> / S <sub>3</sub> |
| AD <sub>12</sub> ↔ 4 | 37 ↔ AD <sub>17</sub> / S <sub>4</sub> |
| AD <sub>11</sub> ↔ 5 | 36 ↔ AD <sub>18</sub> / S <sub>5</sub> |
| AD <sub>10</sub> ↔ 6 | 35 ↔ AD <sub>19</sub> / S <sub>6</sub> |
| AD <sub>9</sub> ↔ 7  | 34 ↔ BHE/ S <sub>7</sub>               |
| AD <sub>8</sub> ↔ 8  | 33 ← MN/ $\overline{\text{MX}}$        |
| AD <sub>7</sub> ↔ 9  | 32 → $\overline{\text{RD}}$            |
| AD <sub>6</sub> ↔ 10 | 31 ← HOLD                              |
| AD <sub>5</sub> ↔ 11 | 30 ↔ HLDA                              |
| AD <sub>4</sub> ↔ 12 | 29 → $\overline{\text{WR}}$            |
| AD <sub>3</sub> ↔ 13 | 28 → M/ $\overline{\text{IO}}$         |
| AD <sub>2</sub> ↔ 14 | 27 → DT/ $\overline{\text{R}}$         |
| AD <sub>1</sub> ↔ 15 | 26 → $\overline{\text{DEN}}$           |
| AD <sub>0</sub> ↔ 16 | 25 → ALE                               |
| NMI ↔ 17             | 24 → $\overline{\text{INTA}}$          |
| INTR → 18            | 23 ← $\overline{\text{TEST}}$          |
| CLK → 19             | 22 ← READY                             |
| GND ← 20             | 21 ← RESET                             |

# Basic Architecture of 8086 and 8088 Microprocessors

## Architecture



### Execution Unit (EU)

EU executes instructions that have already been fetched by the BIU.

BIU and EU functions separately.

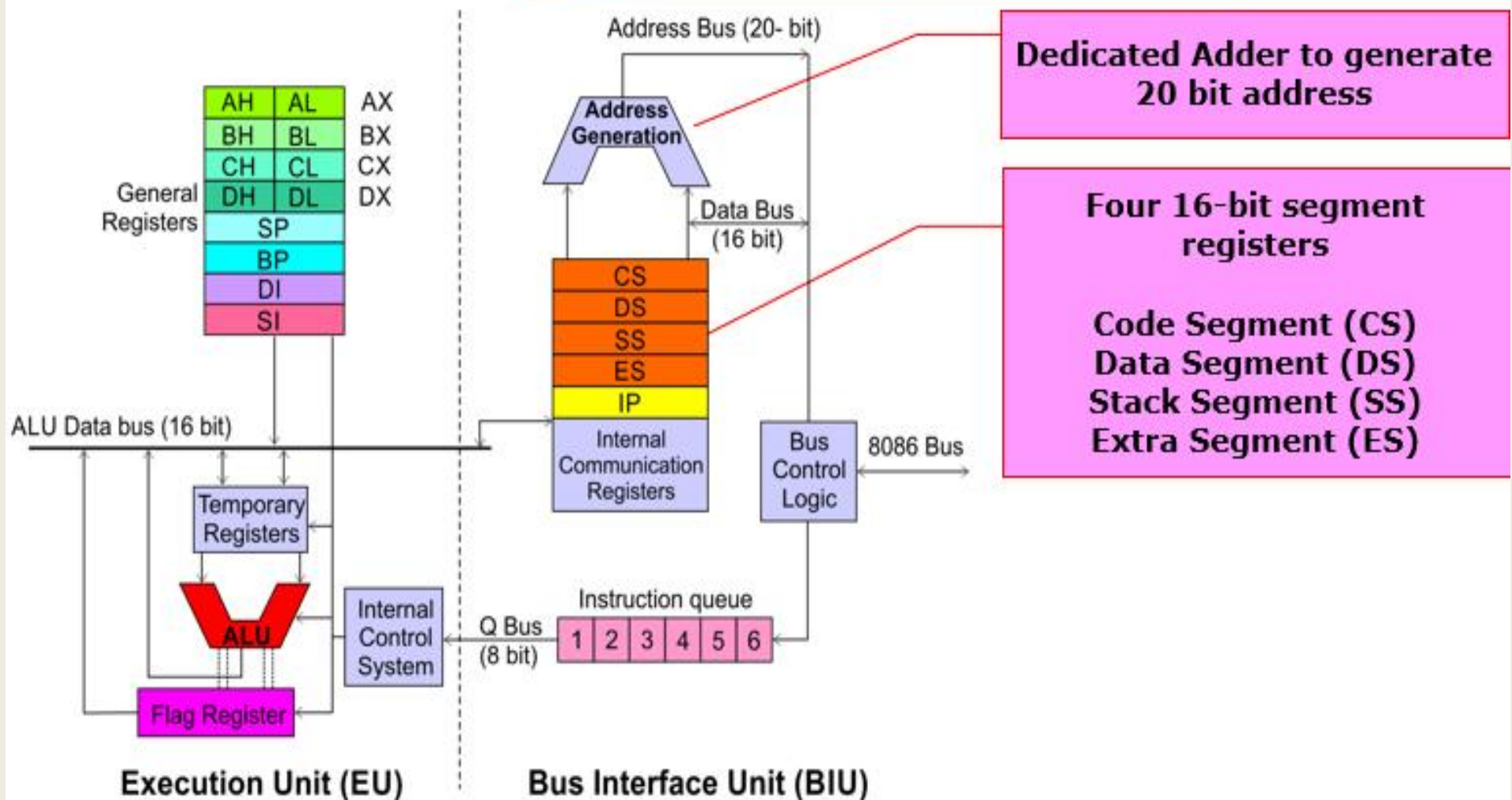
### Bus Interface Unit (BIU)

BIU fetches instructions, reads data from memory and I/O ports, writes data to memory and I/O ports.

# Basic Architecture of 8086 and 8088 Microprocessors

## Architecture

## Bus Interface Unit (BIU)

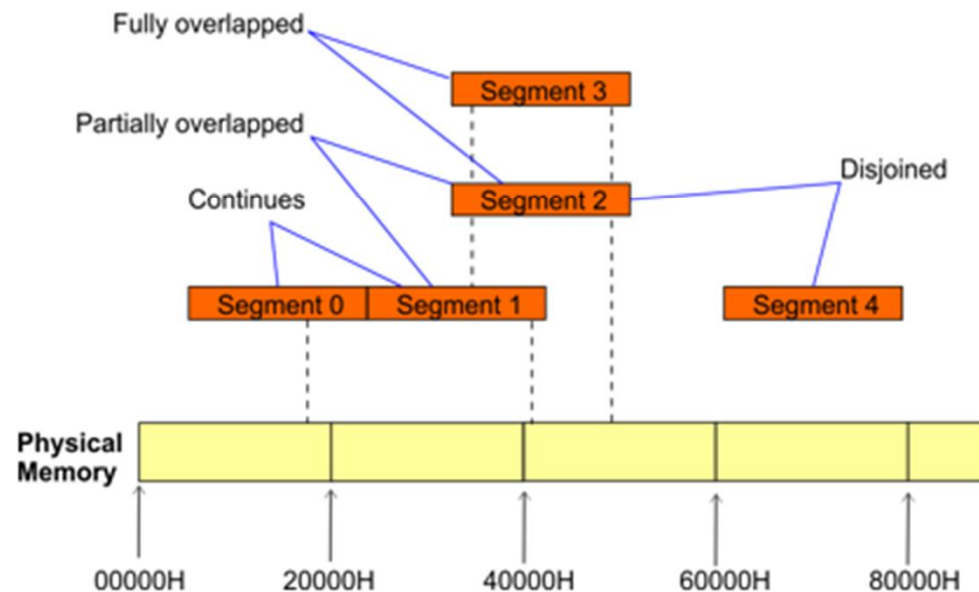
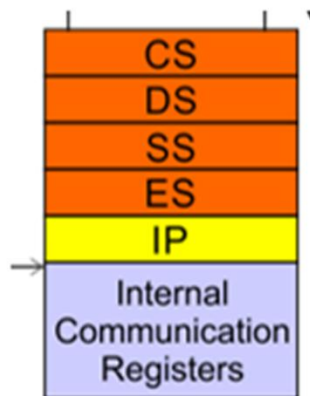


# Basic Architecture of 8086 and 8088 Microprocessors

## Architecture

## Bus Interface Unit (BIU)

### Segment Registers



- 8086's 1-megabyte memory is divided into segments of up to 64K bytes each.
- The 8086 can directly address four segments (256 K bytes within the 1 M byte of memory) at a particular time.
- Programs obtain access to code and data in the segments by changing the segment register content to point to the desired segments.



# Basic Architecture of 8086 and 8088 Microprocessors

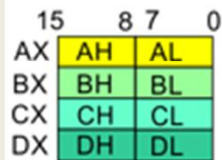
## Architecture

## Bus Interface Unit (BIU)

### Segment Registers

### Code Segment Register

- 16-bit
- CS contains the base or start of the current code segment; IP contains the distance or offset from this address to the next instruction byte to be fetched.
- BIU computes the 20-bit physical address by logically shifting the contents of CS 4-bits to the left and then adding the 16-bit contents of IP.
- That is, all instructions of a program are relative to the contents of the CS register multiplied by 16 and then offset is added provided by the IP.



EU



BIU

# Basic Architecture of 8086 and 8088 Microprocessors



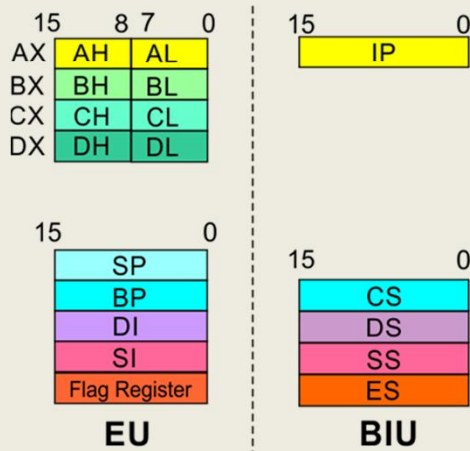
## Architecture

## Bus Interface Unit (BIU)

### Segment Registers

### Data Segment Register

- 16-bit
- Points to the current data segment; operands for most instructions are fetched from this segment.
- The 16-bit contents of the Source Index (SI) or Destination Index (DI) or a 16-bit displacement are used as offset for computing the 20-bit physical address.





# Basic Architecture of 8086 and 8088 Microprocessors

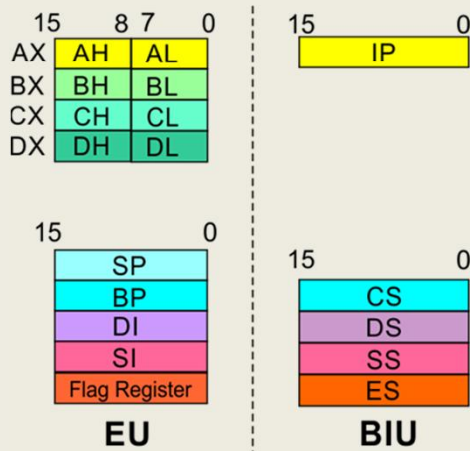
## Architecture

## Bus Interface Unit (BIU)

### Segment Registers

### Stack Segment Register

- 16-bit
- Points to the current stack.
- The 20-bit physical stack address is calculated from the Stack Segment (SS) and the Stack Pointer (SP) for stack instructions such as **PUSH** and **POP**.
- In based addressing mode, the 20-bit physical stack address is calculated from the Stack segment (SS) and the Base Pointer (BP).



# Basic Architecture of 8086 and 8088 Microprocessors



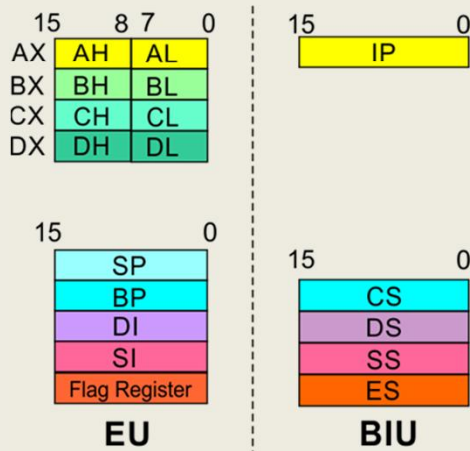
## Architecture

## Bus Interface Unit (BIU)

### Segment Registers

### Extra Segment Register

- 16-bit
- Points to the extra segment in which data (in excess of 64K pointed to by the DS) is stored.
- String instructions use the ES and DI to determine the 20-bit physical address for the destination.



# Basic Architecture of 8086 and 8088 Microprocessors



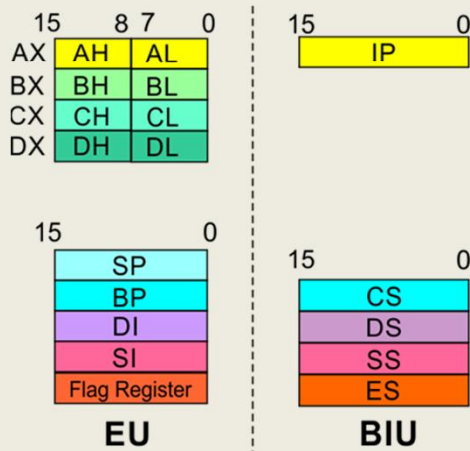
## Architecture

## Bus Interface Unit (BIU)

### Segment Registers

### Instruction Pointer

- 16-bit
- Always points to the next instruction to be executed within the currently executing code segment.
- So, this register contains the 16-bit offset address pointing to the next instruction code within the 64Kb of the code segment area.
- Its content is automatically incremented as the execution of the next instruction takes place.

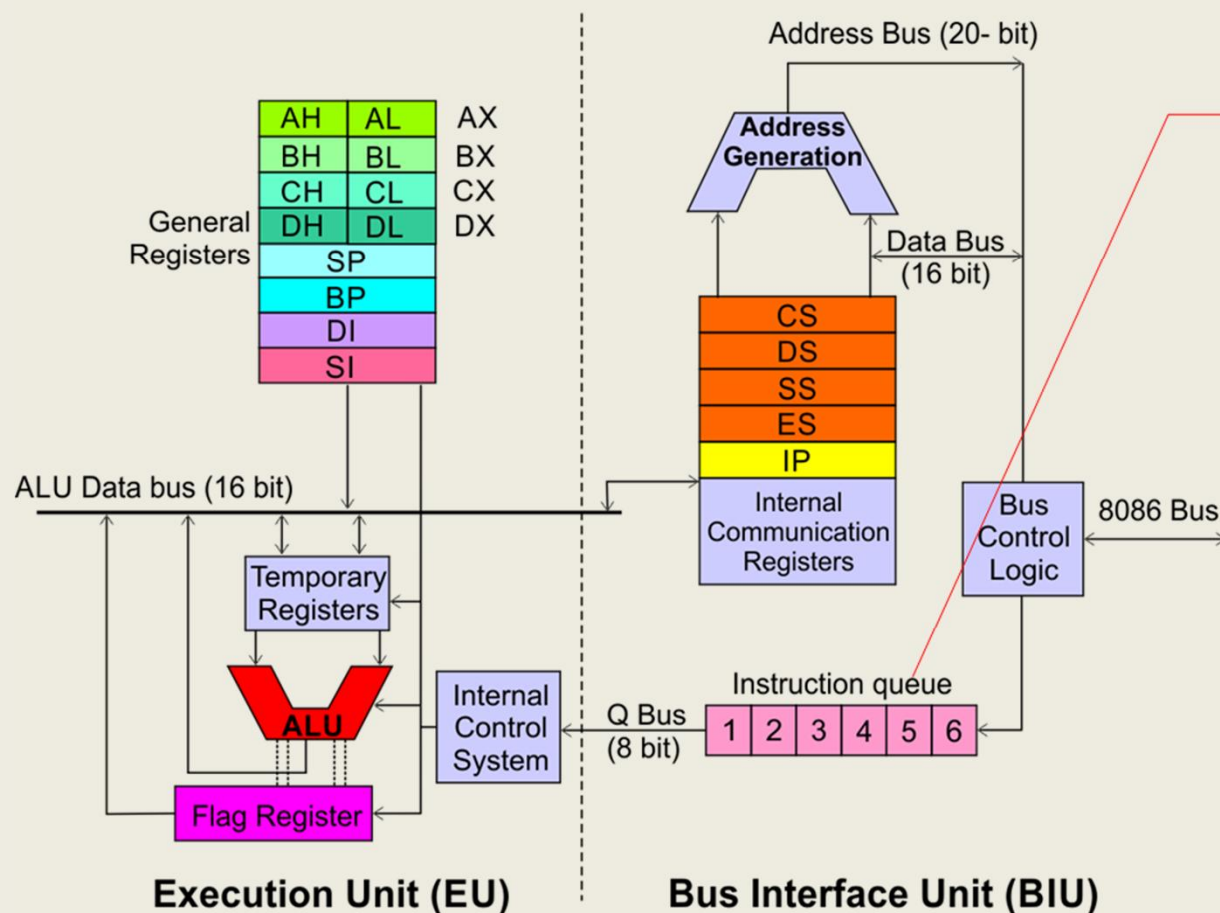




# Basic Architecture of 8086 and 8088 Microprocessors

## Architecture

## Bus Interface Unit (BIU)



### Instruction queue

- A group of First-In-First-Out (FIFO) in which up to 6 bytes of instruction code are pre fetched from the memory ahead of time.
- This is done in order to speed up the execution by overlapping instruction fetch with execution.
- This mechanism is known as pipelining.

# Basic Architecture of 8086 and 8088 Microprocessors

## Architecture

## Execution Unit (EU)

**EU decodes and executes instructions.**

**A decoder in the EU control system translates instructions.**

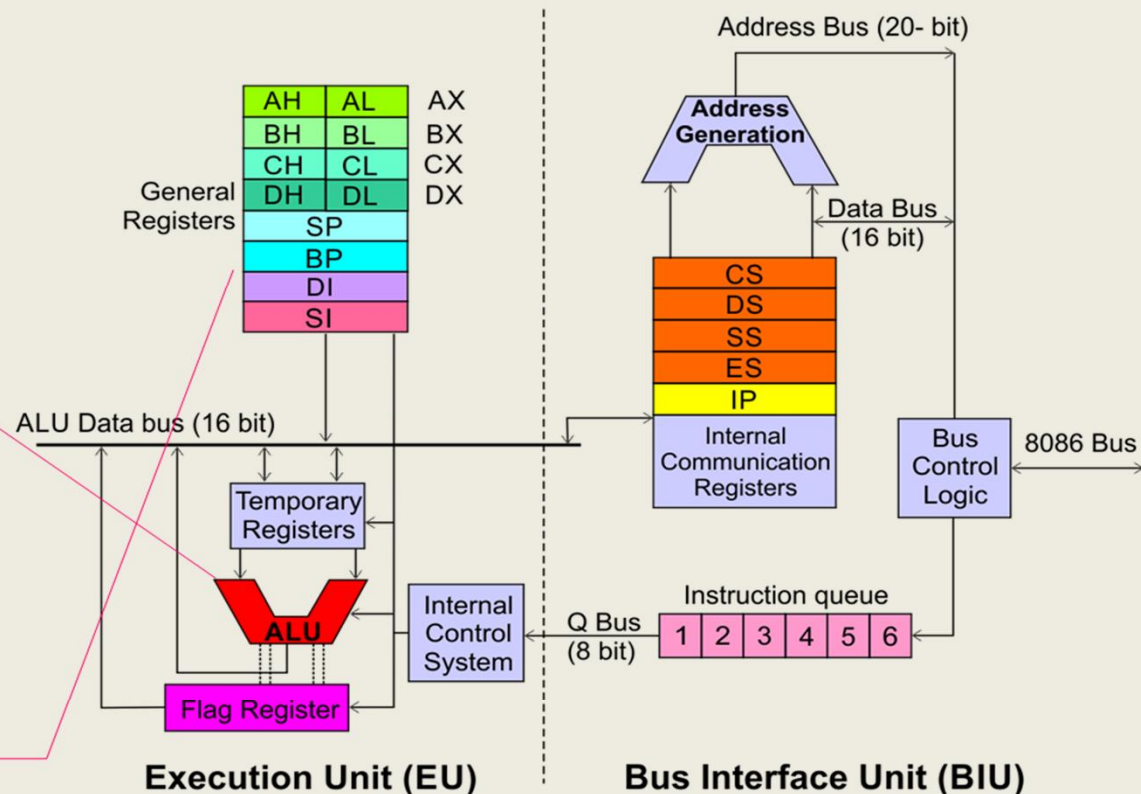
**16-bit ALU for performing arithmetic and logic operation**

**Four general purpose registers (AX, BX, CX, DX);**

**Pointer registers (Stack Pointer, Base Pointer);**

**and**

**Index registers (Source Index, Destination Index) each of 16-bits**



**Some of the 16 bit registers can be used as two 8 bit registers as :**

**AX can be used as AH and AL  
BX can be used as BH and BL  
CX can be used as CH and CL  
DX can be used as DH and DL**

# Basic Architecture of 8086 and 8088 Microprocessors



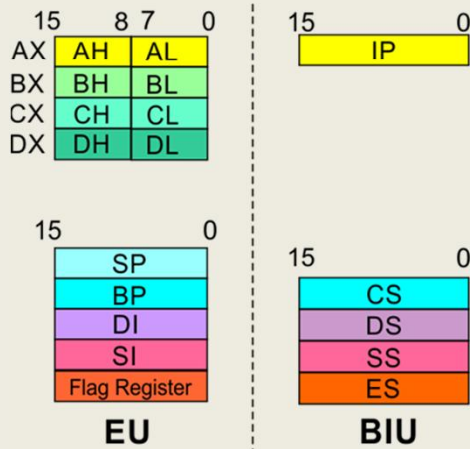
## Architecture

## Execution Unit (EU)

### EU Registers

### Accumulator Register (AX)

- Consists of two 8-bit registers AL and AH, which can be combined together and used as a 16-bit register AX.
- AL in this case contains the low order byte of the word, and AH contains the high-order byte.
- The I/O instructions use the AX or AL for inputting / outputting 16 or 8 bit data to or from an I/O port.
- Multiplication and Division instructions also use the AX or AL.



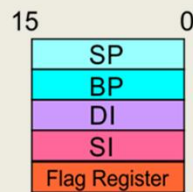
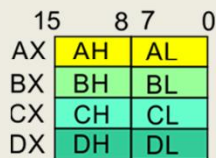
# Basic Architecture of 8086 and 8088 Microprocessors



## Architecture

## Execution Unit (EU)

### EU Registers



**EU**

### Base Register (BX)

- Consists of two 8-bit registers BL and BH, which can be combined together and used as a 16-bit register BX.
- BL in this case contains the low-order byte of the word, and BH contains the high-order byte.
- This is the only general purpose register whose contents can be used for addressing the 8086 memory.
- All memory references utilizing this register content for addressing use DS as the default segment register.



**BIU**



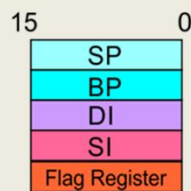
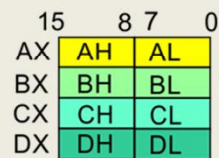
# Basic Architecture of 8086 and 8088 Microprocessors

8086 Microprocessor

## Architecture

## Execution Unit (EU)

### EU Registers



EU



BIU

### Counter Register (CX)

- Consists of two 8-bit registers CL and CH, which can be combined together and used as a 16-bit register CX.
- When combined, CL register contains the low order byte of the word, and CH contains the high-order byte.
- Instructions such as **SHIFT**, **ROTATE** and **LOOP** use the contents of CX as a counter.

### Example:

The instruction **LOOP START** automatically decrements CX by 1 without affecting flags and will check if [CX] = 0.

If it is zero, 8086 executes the next instruction; otherwise the 8086 branches to the label **START**.

# Basic Architecture of 8086 and 8088 Microprocessors



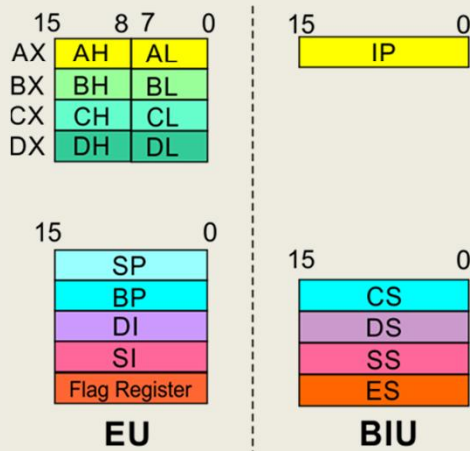
## Architecture

## Execution Unit (EU)

### EU Registers

### Data Register (DX)

- Consists of two 8-bit registers DL and DH, which can be combined together and used as a 16-bit register DX.
- When combined, DL register contains the low order byte of the word, and DH contains the high-order byte.
- Used to hold the high 16-bit result (data) in 16 X 16 multiplication or the high 16-bit dividend (data) before a  $32 \div 16$  division and the 16-bit remainder after division.



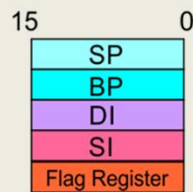
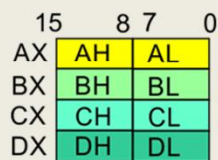
# Basic Architecture of 8086 and 8088 Microprocessors



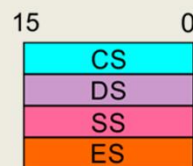
## Architecture

## Execution Unit (EU)

### EU Registers



**EU**



**BIU**

### Stack Pointer (SP) and Base Pointer (BP)

- SP and BP are used to access data in the stack segment.
- SP is used as an offset from the current SS during execution of instructions that involve the stack segment in the external memory.
- SP contents are automatically updated (incremented/decremented) due to execution of a POP or PUSH instruction.
- BP contains an offset address in the current SS, which is used by instructions utilizing the based addressing mode.

# Basic Architecture of 8086 and 8088 Microprocessors



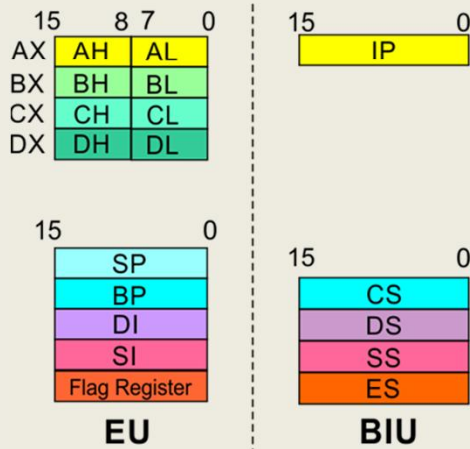
## Architecture

## Execution Unit (EU)

### EU Registers

### Source Index (SI) and Destination Index (DI)

- Used in indexed addressing.
- Instructions that process data strings use the SI and DI registers together with DS and ES respectively in order to distinguish between the source and destination addresses.



# Basic Architecture of 8086 and 8088 Microprocessors



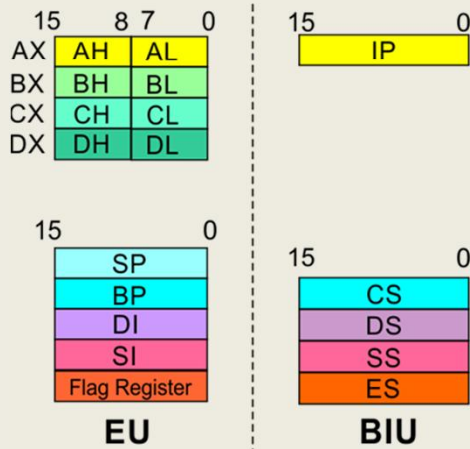
## Architecture

## Execution Unit (EU)

### EU Registers

### Source Index (SI) and Destination Index (DI)

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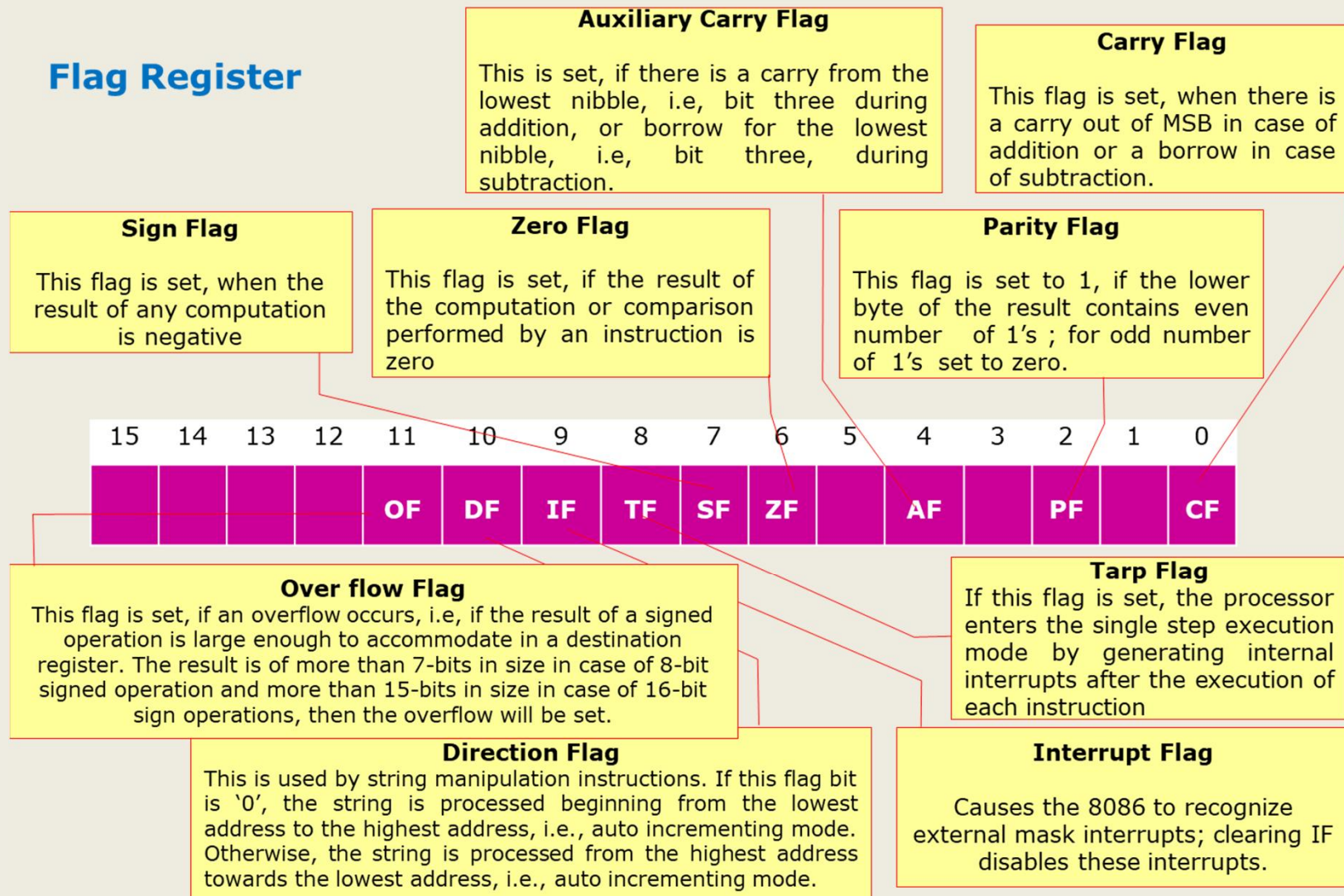


# Basic Architecture of 8086 and 8088 Microprocessors

## Architecture

## Execution Unit (EU)

### Flag Register





# Basic Architecture of 8086 and 8088 Microprocessors

✧ Examples: the status of all status flags after performing the addition and subtraction operation

| Addition                |  |  |  | Subtraction             |  |  |  |
|-------------------------|--|--|--|-------------------------|--|--|--|
| 0110 0101 1101 0001     |  |  |  | 0110 0101 1101 0001     |  |  |  |
| + 0010 0011 0101 1001   |  |  |  | - 0010 0011 0101 1001   |  |  |  |
| <hr/>                   |  |  |  | <hr/>                   |  |  |  |
| 1000 1001 0010 1010     |  |  |  | 1000 1001 0010 1010     |  |  |  |
| SF = 0, ZF = 0, PF = 1, |  |  |  | SF = 1, ZF = 0, PF = 1, |  |  |  |
| CF = 0, AF = 1, OF = 0  |  |  |  | CF = 0, AF = 0, OF = 1  |  |  |  |



# Basic Architecture of 8086 and 8088 Microprocessors

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✧ Exercise 1: Show the status of all status flags after performing the following operation

MOV A 2Bh (load 2BH in register A)

MOV B 19h (load 39H in register B)

ADD B

✧ Exercise 2: Show the status of all status flags after performing the following operation

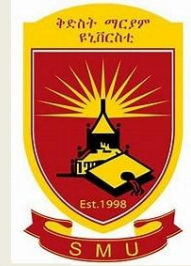
MOV A 2Bh (load 2BH in register A)

MOV B 19h (load 39H in register B)

SUB B

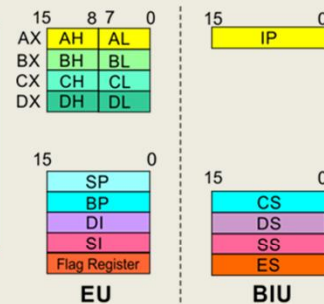
✧ AL = , CF = , PF = , AF = , ZF = , SF = , OF = , ?

# Basic Architecture of 8086 and 8088 Microprocessors



## Architecture

**8086 registers categorized into 4 groups**



| Sl.No. | Type                     | Register width | Name of register               |
|--------|--------------------------|----------------|--------------------------------|
| 1      | General purpose register | 16 bit         | AX, BX, CX, DX                 |
|        |                          | 8 bit          | AL, AH, BL, BH, CL, CH, DL, DH |
| 2      | Pointer register         | 16 bit         | SP, BP                         |
| 3      | Index register           | 16 bit         | SI, DI                         |
| 4      | Instruction Pointer      | 16 bit         | IP                             |
| 5      | Segment register         | 16 bit         | CS, DS, SS, ES                 |
| 6      | Flag (PSW)               | 16 bit         | Flag register                  |



# Basic Architecture of 8086 and 8088 Microprocessors

| Register  | Name of the Register      | Special Function                                                                                         |
|-----------|---------------------------|----------------------------------------------------------------------------------------------------------|
| <b>AX</b> | <b>16-bit Accumulator</b> | <b>Stores the 16-bit results of arithmetic and logic operations</b>                                      |
| <b>AL</b> | <b>8-bit Accumulator</b>  | <b>Stores the 8-bit results of arithmetic and logic operations</b>                                       |
| <b>BX</b> | <b>Base register</b>      | <b>Used to hold base value in base addressing mode to access memory data</b>                             |
| <b>CX</b> | <b>Count Register</b>     | <b>Used to hold the count value in SHIFT, ROTATE and LOOP instructions</b>                               |
| <b>DX</b> | <b>Data Register</b>      | <b>Used to hold data for multiplication and division operations</b>                                      |
| <b>SP</b> | <b>Stack Pointer</b>      | <b>Used to hold the offset address of top stack memory</b>                                               |
| <b>BP</b> | <b>Base Pointer</b>       | <b>Used to hold the base value in base addressing using SS register to access data from stack memory</b> |
| <b>SI</b> | <b>Source Index</b>       | <b>Used to hold index value of source operand (data) for string instructions</b>                         |
| <b>DI</b> | <b>Data Index</b>         | <b>Used to hold the index value of destination operand (data) for string operations</b>                  |

# Basic Architecture of 8086 and 8088 Microprocessors

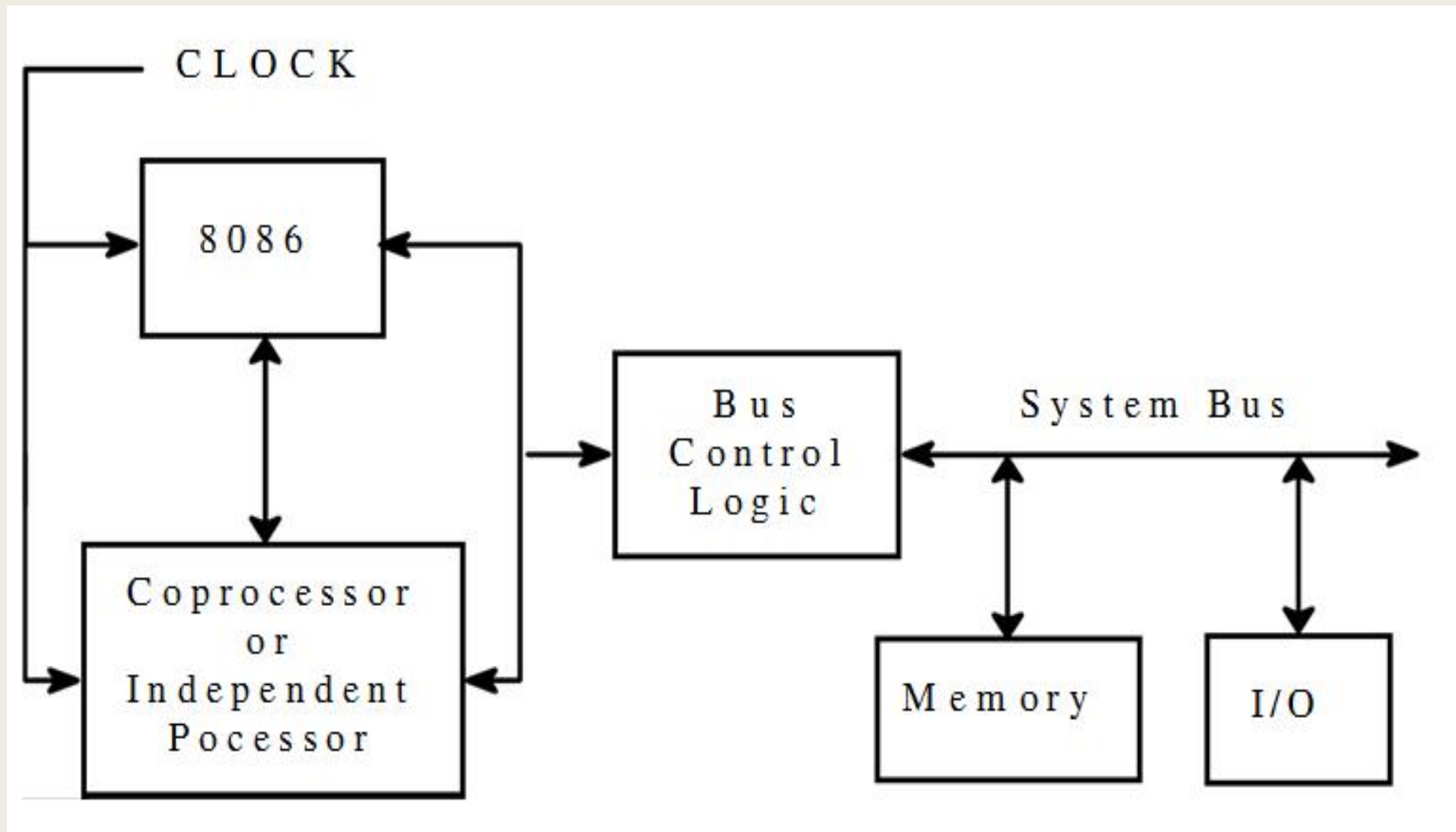


## ✧ *Coprocessor and Multiprocessor configuration*

- Multiprocessor Systems refer to the use of multiple processors that executes instructions simultaneously and communicate with each other using mail boxes and Semaphores.
- Maximum mode of 8086 is designed to implement 3 basic multiprocessor configurations:
  - Coprocessor (8087)
  - Closely coupled (8089)
  - Loosely coupled (Multibus)
- Coprocessors and Closely coupled configurations are similar in that both the 8086 and the external processor shares the:
  - Memory
  - I/O system
  - Bus & bus control logic
  - Clock generator

# Basic Architecture of 8086 and 8088 Microprocessors

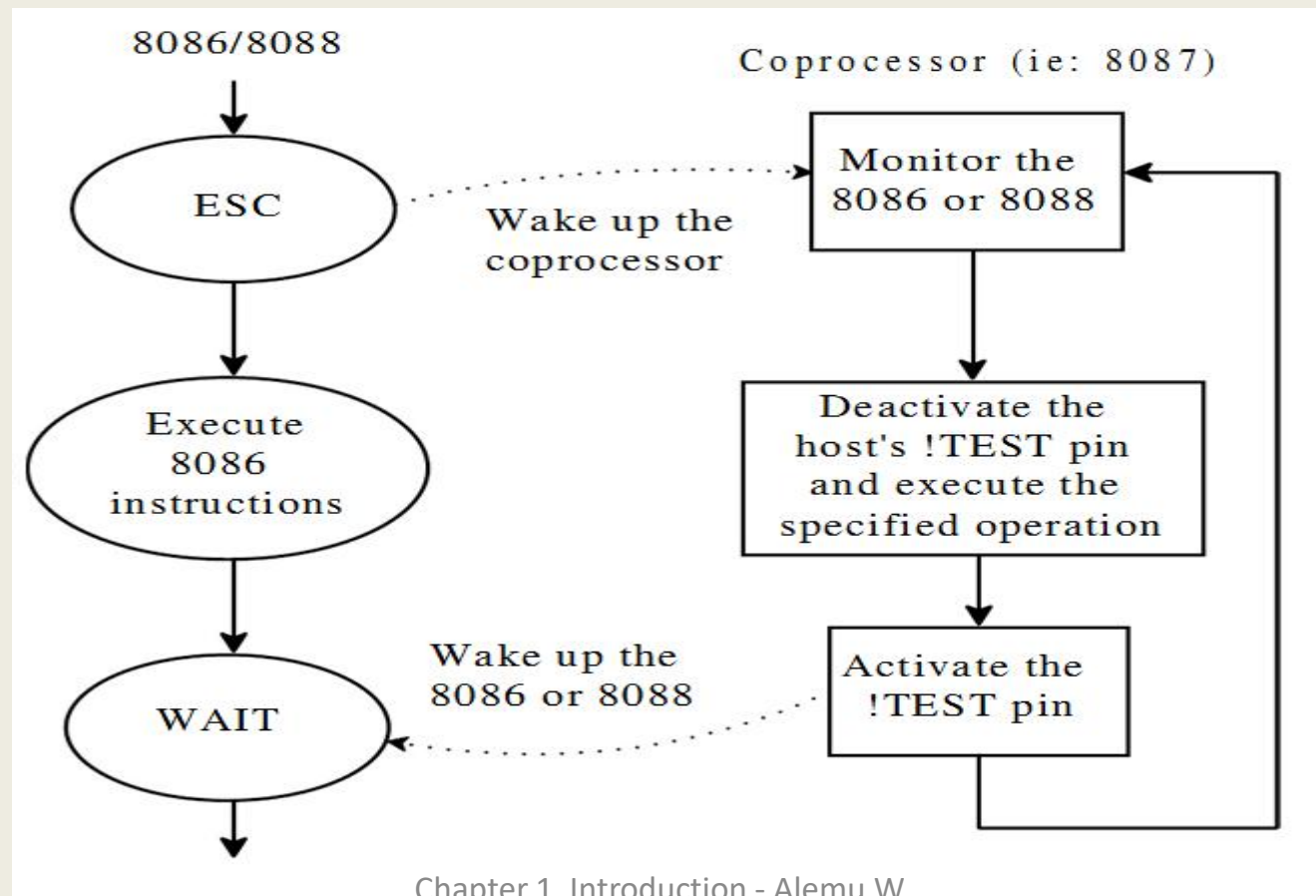
## Coprocessor / Closely Coupled Configuration



# Basic Architecture of 8086 and 8088 Microprocessors

## Coprocessor Execution Example

- ✧ Coprocessor cannot take control of the bus, it does everything through the CPU





# Advantages of Multiprocessor Configuration

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- ✧ High system throughput can be achieved by having more than one CPU.
- ✧ The system can be expanded in modular form.
  - Each bus master module is an independent unit and normally resides on a separate PC board.
  - One can be added or removed without affecting the others in the system.
- ✧ A failure in one module normally does not affect the breakdown of the entire system and the faulty module can be easily detected and replaced
- ✧ Each bus master has its own local bus to access dedicated memory or IO devices. So a greater degree of parallel processing can be achieved.